# Packet Tracer - Inter-VLAN Routing Challenge

## Addressing Table

| Device | Interface | IP Address | Subnet Mask | Default Gateway |
|---|---|---|---|---|
| R1 | G0/0/0 | 172.17.25.2 | 255.255.255.252 | N/A |
| | G0/0/1.10 | 172.17.10.1 | 255.255.255.0 | |
| | G0/0/1.20 | 172.17.20.1 | 255.255.255.0 | |
| | G0/0/1.30 | 172.17.30.1 | 255.255.255.0 | |
| | G0/0/1.88 | 172.17.88.1 | 255.255.255.0 | |
| | G0/0/1.99 | 172.17.99.1 | 255.255.255.0 | |
| S1 | VLAN 99 | 172.17.99.10 | 255.255.255.0 | 172.17.99.1 |
| PC1 | NIC | 172.17.10.21 | 255.255.255.0 | 172.17.10.1 |
| PC2 | NIC | 172.17.20.22 | 255.255.255.0 | 172.17.20.1 |
| PC3 | NIC | 172.17.30.23 | 255.255.255.0 | 172.17.30.1 |
| Server | NIC | 172.17.50.254 | 255.255.255.0 | 172.17.50.1 |

## VLAN and Port Assignments Table

| VLAN | Name | Interface |
|---|---|---|
| 10 | Faculty/Staff | G1/0/11-17 |
| 20 | Students | G1/0/18-24 |
| 30 | Guest(Default) | G1/0/6-10 |
| 88 | Native | G1/0/1 |
| 99 | Management | VLAN 99 |

## Scenario

In this activity, you will demonstrate and reinforce your ability to implement inter-VLAN routing, including configuring IP addresses, VLANs, trunking, and subinterfaces.

## Instructions

Configure the devices to meet the following requirements.

- Assign IP addressing to R1 and S1 based on the Addressing Table.
- Configure the default gateway on S1.

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- Create, name, and assign VLANs on S1 based on the VLAN and Port Assignments Table. Ports should be in access mode. Your VLAN names should match the names in the table exactly.
- Configure G1/0/1 of S1 as a static trunk and assign the native VLAN.
- **All** ports that are not assigned to a VLAN should be disabled.
- Configure inter-VLAN routing on R1 based on the Addressing Table.
- Verify connectivity. R1, S1, and all PCs should be able to ping each other and the server.

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